Biology All In One (BAIO) Learning System

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Chapter 9: Biotechnology – Principles and Processes

Worksheet · Restriction Enzymes & Gel Electrophoresis

I. Multiple Choice Questions

1. How many essential tools of Recombinant DNA Technology (RDT) are listed in the chapter?

(A) 3
(B) 4
(C) 5
(D) 6

2. Which enzyme cuts DNA at specific internal positions?

(A) Exonuclease
(B) Endonuclease
(C) Ligase
(D) Polymerase

3. Which enzyme removes nucleotides from the ends of DNA?

(A) Endonuclease
(B) Exonuclease
(C) Ligase
(D) Restriction enzyme

4. Approximately how many restriction enzymes have been isolated to date?

(A) Over 900
(B) Exactly 50
(C) Only 10
(D) Over 10,000

5. From approximately how many bacterial strains have restriction enzymes been isolated?

(A) 230
(B) 10
(C) 1000
(D) 5

II. Fill in the Blanks

1. Restriction enzymes cut DNA at a specific _______ sequence.

2. Recognition sequences for restriction enzymes are _______ — they read the same in the 5'→3' direction on both strands.

3. The single-stranded overhangs generated by an off-centre cut are called _______.

4. In the enzyme name EcoRI, the first letter denotes the _______ of the source organism.

5. DNA fragments generated by restriction digestion are separated using _______.

6. In gel electrophoresis, DNA fragments migrate toward the _______ because DNA is negatively charged.

III. Match the Following

Column A	Column B
1. Endonuclease	a. Removes nucleotides from DNA ends
2. Exonuclease	b. Stain used to visualise DNA under UV
3. Sticky end	c. Cuts DNA at internal specific positions
4. Ethidium bromide	d. Complementary single-stranded overhang

IV. True / False

1. Restriction enzymes cut DNA exactly at the centre of the palindromic site, always producing blunt ends. [True / False]

2. Over 900 restriction enzymes have been isolated from over 230 bacterial strains. [True / False]

3. DNA ligase is used to join complementary sticky ends together. [True / False]

4. In gel electrophoresis, larger DNA fragments move farther than smaller ones. [True / False]

5. Ethidium bromide combined with UV light is used to visualise DNA bands. [True / False]

V. 2 Marks Questions

1. Explain why restriction enzymes are often called 'molecular scissors'.

2. Describe how sticky ends are generated and why they are useful in constructing recombinant DNA.

VI. 3 Marks Questions

1. Explain the naming convention of a restriction enzyme using EcoRI as an example.

2. Why do smaller DNA fragments travel farther than larger ones in agarose gel electrophoresis?

VII. 5 Marks Questions

1. Explain what restriction enzymes are and how they recognise and cut palindromic DNA sequences. With a diagram, show how EcoRI produces sticky ends and how these help in constructing recombinant DNA.